Notation: $\alpha_{ij} = \varepsilon_i - \varepsilon_j \in \mathbb{Z}^\ell$

$R = \{\alpha_{ij} : 1 \le i,j \le \ell,\ i \ne j\}$ (= the set of all roots for GL_ℓ)

$R_+ = \{\alpha_{ij} : 1 \le i < j \le \ell\}$ (= the set of all positive roots for GL_ℓ)

Definition 3.4.1: Given
- weight $\lambda \in \mathbb{Z}^\ell$
- $R_q, R_t, R_{qt} \subseteq R_+$

Define the corresponding Catalanimal of length ℓ to be the symmetric rational function in the variables $z_1, \dots, z_\ell$

$$H(R_q, R_t, R_{qt}, \lambda) := \sum_{w \in S_\ell} w\left(\frac{z^\lambda \prod_{\alpha \in R_{qt}} (1 - qt z^\alpha)}{\prod_{\alpha \in R_+} (1 - z^{-\alpha}) \prod_{\alpha \in R_q} (1 - q z^\alpha) \prod_{\alpha \in R_t} (1 - t z^\alpha)} \right).$$

in the variables $z_1, \dots, z_\ell$.

Notation:
- $$\Phi(R_q, R_t, R_{qt}, \lambda) := \frac{z^\lambda \prod_{\alpha \in R_+ \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R_+ \setminus R_t} (1 - t z^\alpha)}{\prod_{\alpha \in R_+ \setminus R_{qt}} (1 - qt z^\alpha)}$$

- $$g(R_q, R_t, R_{qt}, \lambda) := \sum_{w \in S_\ell} w\left(z^\lambda \prod_{\alpha \in R_+} (1 - z^\alpha) \prod_{\alpha \in R \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R \setminus R_t} (1 - t z^\alpha) \prod_{\alpha \in R_{qt}} (1 - qt z^\alpha) \right)$$

Then
- $\sigma_{\hat{\rho}}(\Phi(R_q, R_t, R_{qt}, \lambda)) = H(R_q, R_t, R_{qt}, \lambda)$
- $\sigma_{\check{\rho}}(\Phi(R_q, R_t, R_{qt}, \lambda)) = g(R_q, R_t, R_{qt}, \lambda)$

Proof: $$\sigma_{\hat{\rho}}(\Phi(R_q, R_t, R_{qt}, \lambda)) = \sum_{w \in S_\ell} w\left(\Phi(R_q, R_t, R_{qt}, \lambda) \cdot \prod_{i<j} \hat{r}(z_i, z_j) \right)$$

$$= \sum_{w \in S_\ell} w\left(\frac{z^\lambda \prod_{\alpha \in R_+ \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R_+ \setminus R_t} (1 - t z^\alpha)}{\prod_{\alpha \in R_+ \setminus R_{qt}} (1 - qt z^\alpha)} \cdot \prod_{i<j} \frac{1 - qt \frac{z_i}{z_j}}{(1 - \frac{z_j}{z_i})(1 - q \frac{z_i}{z_j})(1 - t \frac{z_i}{z_j})} \right)$$

$$= \sum_{w \in S_\ell} w\left(\frac{z^\lambda \prod_{\alpha \in R_+ \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R_+ \setminus R_t} (1 - t z^\alpha)}{\prod_{\alpha \in R_+ \setminus R_{qt}} (1 - qt z^\alpha)} \cdot \prod_{\alpha \in R_+} \frac{1 - qt z^\alpha}{(1 - z^\alpha)(1 - q z^\alpha)(1 - t z^\alpha)} \right)$$

$$= \sum_{w \in S_\ell} w\left(\frac{z^\lambda \prod_{\alpha \in R_{qt}} (1 - qt z^\alpha)}{\prod_{\alpha \in R_+} (1 - z^{-\alpha}) \prod_{\alpha \in R_q} (1 - q z^\alpha) \prod_{\alpha \in R_t} (1 - t z^\alpha)} \right) = H(R_q, R_t, R_{qt}, \lambda).$$

$$\sigma_{\check{\rho}}(\Phi(R_q, R_t, R_{qt}, \lambda)) = \sum_{w \in S_\ell} w\left(\Phi(R_q, R_t, R_{qt}, \lambda) \cdot \prod_{i<j} \check{r}(z_i, z_j) \right)$$

$$= \sum_{w \in S_\ell} w\left(\frac{z^\lambda \prod_{\alpha \in R_+ \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R_+ \setminus R_t} (1 - t z^\alpha)}{\prod_{\alpha \in R_+ \setminus R_{qt}} (1 - qt z^\alpha)} \cdot \prod_{i<j} (1 - \frac{z_i}{z_j})(1 - q \frac{z_j}{z_i})(1 - t \frac{z_j}{z_i})(1 - qt \frac{z_i}{z_j}) \right)$$

$$= \sum_{w \in S_\ell} w\left(\frac{z^\lambda \prod_{\alpha \in R_+ \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R_+ \setminus R_t} (1 - t z^\alpha)}{\prod_{\alpha \in R_+ \setminus R_{qt}} (1 - qt z^\alpha)} \cdot \prod_{\alpha \in R_+} (1 - z^\alpha)(1 - q z^{-\alpha})(1 - t z^{-\alpha})(1 - qt z^\alpha) \right)$$

$$= \sum_{w \in S_\ell} w\left(z^\lambda \prod_{\alpha \in R_+} (1 - z^\alpha) \prod_{\alpha \in R_{qt}} (1 - qt z^\alpha) \prod_{\alpha \in R_+ \setminus R_q} (1 - q z^\alpha) \cdot \prod_{\alpha \in R_+} (1 - q z^{-\alpha}) \cdot \prod_{\alpha \in R_+ \setminus R_t} (1 - t z^\alpha) \prod_{\alpha \in R_t} (1 - t z^{-\alpha}) \right)$$

$$= \sum_{w \in S_\ell} w\left(z^\lambda \prod_{\alpha \in R_+} (1 - z^\alpha) \prod_{\alpha \in R_{qt}} (1 - qt z^\alpha) \prod_{\alpha \in R \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R \setminus R_t} (1 - t z^\alpha) \right)$$

$$= g(R_q, R_t, R_{qt}, \lambda).$$

e.g. $\ell = 3$, $R_q = \{\alpha_{12}, \alpha_{13}\}$, $R_t = \{\alpha_{13}, \alpha_{23}\}$, $R_{qt} = \{\alpha_{13}\}$, $\lambda = 211$

$$H(R_q, R_t, R_{qt}, (2,1,1)) = \sum_{w \in S_3} w\left(\frac{z_1^2 z_2 z_3 \left(1 - qt\frac{z_1}{z_3}\right)}{\left(1-\frac{z_2}{z_1}\right)\left(1-\frac{z_3}{z_1}\right)\left(1-\frac{z_3}{z_2}\right)\left(1-q\frac{z_1}{z_2}\right)\left(1-q\frac{z_1}{z_3}\right)\left(1-t\frac{z_2}{z_3}\right)\left(1-t\frac{z_2}{z_3}\right)}\right)$$

$$\phi(R_q, R_t, R_{qt}, (2,1,1)) = \frac{z_1^2 z_2 z_3 \left(1-q\frac{z_1}{z_3}\right)\left(1-t\frac{z_1}{z_2}\right)}{\left(1-qt\frac{z_1}{z_2}\right)\left(1-qt\frac{z_1}{z_3}\right)}$$

$$g(R_q, R_t, R_{qt}, (2,1,1)) = \sum_{w \in S_3} w\left(z_1^2 z_2 z_3 \left(1-\frac{z_1}{z_2}\right)\left(1-\frac{z_1}{z_3}\right)\left(1-\frac{z_2}{z_3}\right)\left(1-q\frac{z_1}{z_3}\right)\left(1-q\frac{z_2}{z_3}\right)\left(1-q\frac{z_2}{z_1}\right)\left(1-q\frac{z_1}{z_2}\right)\left(1-t\frac{z_1}{z_2}\right)\left(1-t\frac{z_2}{z_1}\right)\left(1-t\frac{z_3}{z_1}\right)\left(1-t\frac{z_3}{z_2}\right)\left(1-qt\frac{z_1}{z_3}\right)\right)$$

The following are equivalent:

(i) $H \in S_{\hat{P}}$

(ii) $g \in S_{\check{P}}$

(iii) $\phi \in T^\ell + I^\ell$ ($\therefore \phi$ represents an element in S)

$\because$ $\sigma_{\hat{P}}$ is an isomorphism from $\overset{S}{\shortparallel}_{\hat{P}}$ to $S_{\hat{P}}$ and $H = \sigma_{\hat{P}}(\phi)$

$\therefore$ (i) $\Leftrightarrow$ (iii)

Similarly (ii) $\Leftrightarrow$ (iii) b/c $\sigma_{\check{P}}$ is an isomorphism from $\overset{S}{\shortparallel}_{\check{P}}$ to $S_{\check{P}}$ and $g = \sigma_{\check{P}}(\phi)$.

When these conditions hold, by Theorem 3.2.1, there is a corresponding element of the Schiffmann algebra

$$\zeta = \psi(\phi) \left.\begin{aligned} &= \psi(\sigma_{\hat{P}}^{-1}(H)) = \psi \circ \sigma_{\hat{P}}^{-1}(H) = \psi_{\hat{P}}(H) \\ &= \psi(\sigma_{\check{P}}^{-1}(g)) = \psi \circ \sigma_{\check{P}}^{-1}(g) = \psi_{\check{P}}(g) \end{aligned}\right\} \in \mathcal{E}^+.$$

We want to identify certain H s.t. $H \in S_{\hat{P}}$ and the corresponding $\zeta \in \Lambda(X^{m,n}) \subseteq \mathcal{E}^+$ for some $m > 0$.

i.e. the corresponding Schiffmann algebra for H is $f[-MX^{m,n}] \in \mathcal{E}^+$.

Def 3.4.2: Let $H = H(R_q, R_t, R_{qt}, \lambda)$ be a Catalanimal. If $\psi_{\hat{P}}(H) \in \Lambda(X^{m,n})$ for some $m > 0$, we call the symmetric function f s.t. $\psi_{\hat{P}}(H) = f[-MX^{m,n}]$ its cub, and write $\mathrm{cub}(H) = f$.

Summary: Given $H = H(R_q, R_t, R_{qt}, \lambda)$ of length ℓ s.t. $\psi_{\hat{P}}(H) = f[-MX^{m,n}]$, $m > 0$, we have

$$\begin{tikzcd}
 & & H \in S_{\hat{P}}^\ell \arrow[dd, "\psi_{\hat{P}}"] \\
\phi \in T^\ell + I^\ell \arrow[urr, "\sigma_{\hat{P}}^\ell"] \arrow[r, two heads] \arrow[ddrr, "\sigma_{\check{P}}^\ell"'] & S^\ell = (T^\ell + I^\ell)/I^\ell \arrow[ur, "\sigma_{\hat{P}}^\ell"'] \arrow[r, "\psi"] \arrow[d, "\sigma_{\check{P}}^\ell"] & (\mathcal{E}^+)^{(\ell,0)} \ni f[-MX^{m,n}] \\
 & g \in S_{\check{P}}^\ell \arrow[ur, "\psi_{\check{P}}"'] &
\end{tikzcd}$$